

Neha Parepalli

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PROFILE

AI and Data Engineering professional with experience building GenAI workflows, ETL pipelines, data platforms, and AI applications using Python, FastAPI, Docker, SQL, RAG, and cloud technologies. Interested in AI engineering, infrastructure intelligence, and production AI workflows.

SKILLS

Data Engineering: Python, SQL, PySpark, Databricks, Airflow, ETL/ELT, Data Modeling

Generative AI: LLMs, RAG, LangChain, Ollama, ChromaDB, Prompt Engineering, OCR

Backend & Apps: FastAPI, REST APIs, Docker, React, Vite, GitHub Actions, CI/CD

Cloud & Analytics: Azure, Pandas, NumPy, Workflow Automation, Root Cause Analysis

Data Platforms: dbt, DuckDB, Structured Streaming, Parquet

PROFESSIONAL EXPERIENCE

ITERGO Informationstechnologie GmbH

09/2025 – 03/2026

Working Student | Generative AI

Düsseldorf, Germany

- Developed and optimized prompts for **30+ GenAI use cases** using ERGO GPT and enterprise data.
- Analyzed model outputs and refined prompting strategies to improve response quality and relevance.
- Improved operational efficiency by **30%** through AI-driven workflow automation.

ITERGO Informationstechnologie GmbH

03/2025 – 08/2025

AI Intern | Prompt Engineer

Düsseldorf, Germany

- Developed datasets for **20+ AI use cases** supporting enterprise GenAI initiatives.
- Optimized prompting and evaluation workflows for consistent AI-generated outputs.
- Automated testing and validation processes, reducing manual effort by **50%**.

Accenture Services Private Limited

09/2021 – 08/2023

Data Engineering Analyst | Client: Linde PLC, Germany

Mumbai, India

- Developed Python and SQL ETL pipelines processing **1M+ records/month**.
- Built reporting and interface integrations, reducing operational effort by **25%**.
- Implemented validation controls across enterprise data pipelines and workflows.

PROJECTS

AI Incident Copilot – GenAI Incident Intelligence Platform

[GitHub]

- Built an AI-powered platform for automated **root-cause analysis** of production incidents.
- Implemented **RAG-based retrieval**, diagnosis generation, and remediation recommendations.
- Developed the solution using **FastAPI, Ollama, ChromaDB, Docker, and LLMs**.

Document Intelligence Data Platform

[GitHub]

- Built a platform transforming **unstructured documents** into analytics-ready insights.
- Developed **semantic retrieval** using embeddings, FAISS, and Sentence Transformers.
- Implemented **pipeline orchestration**, validation, and analytics workflows using dbt and DuckDB.

EDUCATION

Darmstadt University of Applied Sciences

09/2023 – 03/2026

M.Sc. Electrical Engineering & Information Technology

Darmstadt, Germany

St. Francis Institute of Technology

08/2017 – 06/2021

B.E. Electronics & Telecommunication Engineering | CGPA: 8.61/10

Mumbai, India

LANGUAGES

German (B1), English (C2)